



Proofs and Modern Mathematics

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[Upon hearing that one of his students had dropped out to study poetry:] “Good, he did not have enough imagination to become a mathematician.”

– *David Hilbert*

Course Description and Goals

This course is designed to give students a comprehensive understanding of the foundations of modern mathematics as well as a practical methodology for future study in advanced courses. It uniquely cultivates the ability to observe, analyze, reason, and create. This course prepares students to excel in computational mathematics courses such as multivariable calculus and linear algebra, to have a significant advantage in fields of study that require careful logical thinking like engineering and law, and to be stronger communicators of meaning through music, art, and literature. In the fall semester, students will study mathematical logic, general axiomatic systems, methods of proof, sets, relations, and categories. In the spring semester, students will apply these concepts to an in-depth study of finite groups. Additionally, throughout the year students will read a number of primary and secondary texts exploring the history, philosophy, and theology of mathematics. Upon completion of this course, the student will be able to translate patterns

into the language of sets and functions, explore the categories of sets and finite groups, read and write all kinds of mathematical proofs, and interpret the motivation and purpose of mathematical studies. The student will learn to compose permutations in a variety of forms. The student will be able to prove divisibility relationships of integers, the irrationality of real numbers, set equality, set cardinality, group structure, group equality, and more. Students will also have the wisdom to judge the strength and significance of mathematical theorems and appreciate their beauty through rigorous exploration and explanation.

Textbook

Bloch, Ethan D. *Proofs and Fundamentals: A First Course in Abstract Mathematics*. 2nd edition. In *Undergraduate Texts in Mathematics*. Edited by Sheldon Axler and Kenneth Ribet New York: Springer, 2011.

Excerpts for Supplemental Reading

G.K. Chesterton, *The Two Poets of Saffron Park* from *The Man Who Was Thursday*. Jonathan Edwards, *Treatise on the End for which God Created the World*. G.H. Hardy, *A Mathematician's Apology*. Lewis Carroll, *Alice's Adventure in Wonderland*. Paul Lockhart, *A Mathematician's Lament*. Catherine Pickstock, *Music: Soul, City, and Cosmos after Augustine*. David Eugene Smith, *Religio Mathematici*.

Selected Bibliography

Chartrand, Gary, Albert D. Polimeni, and Ping Zhang. *Mathematical Proofs: A Transition to Advanced Mathematics*. 3rd edition. Boston: Pearson Education, Inc., 2013.

Gerstein, Larry J. *Introduction to Mathematical Structures and Proofs*. 2nd edition. In *Undergraduate Texts in Mathematics*. Edited by Sheldon Axler and Kenneth Ribet. New York: Springer, 2012.

Fraleigh, John. *A First Course in Abstract Algebra*. 6th edition. Reading, Massachusetts: Addison Wesley Longman, Inc, 2002.

Riehl, Emily. *Category Theory in Context*.

Cummings, Jay. *Proofs: A Long-Form Mathematics Textbook*.

Participation and Homework

Homework will be assigned every week. All homework must be submitted in L^AT_EX formatting. Homework will be returned with thorough feedback, and corrections can be made for full marks within a prescribed time frame. Homework will be graded on readability, logical consistency, and mathematical creativity. Late homework will be penalized 10% per day. On the reading portion of the homework, reading means consciously recognizing every word. You are not expected to understand every sentence, argument, or paragraph, but you are asked to consider each section in its entirety, including the exercises. Homework may be submitted electronically via email in pdf format before 6 AM on the date it is due. Otherwise, it should be printed and submitted at the beginning of class.

One paper will be assigned during the fall semester. The student will write a 750-word argument on a subject and thesis of their choice. To complete the project, the student will alter and resubmit the paper according to new instructions. The student will be evaluated on the quality of the argument as represented in each of the drafts.

A second paper will be assigned during the spring semester. The student will write a 500-word essay on his or her personal perspective on the relationship of mathematics to life generally. Should everyone study mathematics? Is mathematics useful? How does mathematics relate to religion and spirituality?

Quizzes and Exams

A 10-minute, 10-question quiz will be given weekly to assess basic comprehension of the vocabulary, theorems, and computations. There will be three exams per semester. The exams will consist entirely of proofs drawn from a list of theorems provided to the students. All of these theorems will have been proved either in class or on the homework. Students will have one full period to complete each exam. As this is a course for seniors, there will be no final exam.

Absences and Make-Up Work

In the case of an absence, it is the student's responsibility to contact the instructor and catch up on announcements and assignments. Refer to the Family Handbook for more information about homework and assessment extensions for excused absences and penalties for unexcused absences. In the special case that two-thirds of the students are absent for 1-2 days, the students present will use the period to practice, review, and/or further explore content already covered. The absent students will not be responsible for any make-up work. If the same students are absent for 3 or more days, they will be assigned reading and homework to complete independently while the class conversation moves forward.

Grading

An "objective" numerical grade will be calculated based on reading and homework sets (30%), quizzes (30%), and exams (40%). Please refer to the Family Handbook for the letter grade scale.

Plagiarism

Plagiarism is any instance of representing an idea of someone else as your own, including material found on the internet even when the author's information is not readily available. Plagiarism is a serious academic offense, even when done unintentionally. Any instance of plagiarism will be referred to the Head of School, and the student will be at risk of receiving no credit on the assignment and potentially failing the course. Using a large language model like ChatGPT to write L^AT_EX code is highly encouraged for formatting; however, sharing finished code with another student will be considered plagiarism. Your homework should be unique and personal at the code level.

Fall Semester Topic Overview

Week 1	Sept. 1	Modern mathematics
Week 2	Sept. 8	Mathematical logic
Week 3	Sept. 15	Modular arithmetic
Week 4	Sept. 22	Divisibility
Week 5	Sept. 29	Irrational numbers
Week 6	Oct. 6	Mathematical induction
Week 7	Oct. 13	Set operations
Week 8	Oct. 20	Power, product, and indexed sets
Week 9	Oct. 27	Measurable sets
Week 10	Nov. 3	Functions
Week 11	Nov. 10	Binary relations
Week 12	Nov. 17	Binary operations
Week 13	Dec. 1	Categories
Week 14	Dec. 8	Morphisms
Week 15	Dec. 15	Functors
Week 16	Jan. 5	Universal Properties

Spring Semester Topic Overview

Week 1	Jan. 19	Finite Groups
Week 2	Jan. 26	Subgroups
Week 3	Feb. 2	Homomorphisms
Week 4	Feb. 9	Automorphisms
Week 5	Feb. 16	Cyclic groups
Week 6	Feb. 23	Generators
Week 7	Mar. 2	Product groups
Week 8	Mar. 9	Normal subgroups
Week 9	Mar. 16	Quotient groups
Week 10	Apr. 6	The Isomorphism Theorems
Week 11	Apr. 13	Classification of Small Groups
Week 12	Apr. 20	Abelianization
Week 13	Apr. 27	Rings and Fields
Week 14	May 4	Ideals
Week 15	May 11	Polynomial Rings
Week 16	May 18	Linear Groups

Proofs and Modern Mathematics Syllabus Signature Page

Sign and submit this page to the instructor by Friday of the first week. Keep the syllabus in your binder for the rest of the year.

“I have read the entire syllabus for Proofs and Modern Mathematics.”

Student Name Printed

Student Signature and Date

Parent or Guardian Name Printed

Parent or Guardian Signature and Date